

Jake Callcut

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TECHNICAL SKILLS

- **Languages:** Python, Java, Rust, Dart, TypeScript
- **ML & Data:** PyTorch, Hugging Face, scikit-learn, Pandas, matplotlib, Jupyter
- **Tools & Infrastructure:** Azure, Docker, Git, Flutter, Socket.IO, Kubernetes

RELEVANT EXPERIENCE

STMicronics

05/2026 – Present

AI & Machine Learning Intern

- Independently researching and implementing state-of-the-art generative AI methods for synthetic data augmentation within the Data and Knowledge Engineering Team, improving defect detection accuracy on semiconductor wafers by 8 percentage points.
- Applying diffusion models with LoRA fine-tuning on proprietary datasets to generate high-fidelity synthetic wafer imagery, that expanded existing training sets by 20%.
- Utilising PyTorch alongside CNN and transformer architectures to evaluate and benchmark data augmentation strategies, establishing synthetic data as the most effective supplement.

Diamond

06/2025 – 03/2026

Contract Software Engineer

- Designed and built the company's consumer mobile app and companion business platform in Flutter, owning delivery end-to-end from UI/UX through deployment.
- Produced technical material for investor pitches and funding applications, contributing to securing a place on the STAC Scale accelerator programme and pre-seed funding of £150,000.

GI Healthcare Industries

08/2024 – 11/2025

Software Engineer (promoted from UI/UX Intern)

- Architected and implemented a multi-machine control system with a custom server-client design using Flutter, TypeScript, and Socket.IO, and a custom protocol ensuring high fault tolerance and taking the company's new product from TRL 3 to TRL 6, supporting a £250,000 pre-seed round.
- Attended stakeholder product demonstrations, engaging with end-users to gather requirements, observe operational needs, and provide feedback to improve system reliability and usability.

EDUCATION

Heriot-Watt University

09/2022 – 06/2027

MEng Software Engineering - On Track for First Class Honours (89/100)

- Three times awarded the Deputy Principal's Award for academic excellence.
- **Relevant Modules:** Data Mining & Machine Learning, Artificial Intelligence & Intelligent Agents, Rigorous Methods for Software Engineering, Big Data Management
- Fund Analyst at the Heriot-Watt Investment Society

PROJECTS & EXTRA-CURRICULAR

Social Sentiment Analysis for Sector Level Stock Market Prediction | Dissertation Project (First Class 90/100)

- Designed and implemented a pipeline scoring 900,000+ social media posts for sentiment data using finBERT and merging aggregated sentiment with returns across 14 securities, validating sentiment signal against VIX ($r = -0.562$).
- Compared and evaluated logistic regression, random forest, and gradient boosting with time series cross-validation, finding cyclical market sectors more predictable (0.675 accuracy) than defensive sectors and commodities (0.350).

Tensor Room | Collaborative AI Project

- Collaborated on building an interior design tool that lets users redesign rooms through natural language, combining Qwen-Image and SAM3 to generate photorealistic edits to real room photos in 20 seconds.
- Used model quantisation and model offloading to allow the pipeline to run locally on <16GB VRAM.
- Implemented complex guardrails to prevent misuse.

Prediction Market Calibration | Personal Research Project

- Built a modular Python pipeline to collect 20,000+ resolved market data from the Kalshi and Polymarket APIs, unifying differing schemas into a single dataset.
- Assessed calibration using reliability curves and Brier scores across 1 hour, 1 day, 1 week, and 1 month horizons, finding both markets calibrated at the 1 day horizon (Kalshi at 0.163, Polymarket at 0.026) with performance degrading at the longer horizons.

ADDITIONAL INFORMATION

- **Certifications:** Bloomberg Market Concepts, MathWorks: Deep Learning.
- **Teaching:** Java Lab Assistant, Heriot-Watt University, freelance mathematics tutor.